North South University

Department of Electrical & Computer Engineering

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Course: CSE440(Artificial Intelligence-AI)

Section: 01

Group No: 07

Project: Develop a Poker game simulator with Expectiminimax for strategic decision-making in uncertain events. Design the environment, necessary components, and evaluate algorithm performance.

Week-1 (10 Feb – 16 Feb):

I read the updated course manual in detail to understand expectations for reports, presentations, GitHub structure, and deadlines. In group meetings I clarified my role, focusing on user- interaction design, documentation, and testing support. I studied simplified Texas Hold'em rules and introductory material on minimax and expectiminimax to follow the technical discussions.

Week-2 (17 Feb – 23 Feb):

I wrote a short requirement note from the human- player perspective, describing the sequence of prompts, available actions (fold, call, raise), and outputs (cards revealed, pot size, round summaries). I sketched a text- based interface layout and suggested clear, concise wording for prompts and results. I reviewed the leader's architecture design and provided feedback to keep the user interface consistent with the internal data structures.

Week-3 (24 Feb – 1 Mar):

Once a basic game engine was running, I executed multiple test games to check that rounds progressed correctly from dealing to showdown. I designed simple scenarios (early fold, everyone calls, multiple raises) and checked whether chip counts and pot updates matched expectations. I recorded any confusing or ambiguous outputs and created a list of issues that needed correction or clearer printing.

Week-4 (2 Mar – 8 Mar):

I helped verify the hand evaluation module by preparing example hands such as flush vs straight and full house vs four- of- a- kind. Using these examples, I checked that the evaluator returns the correct ranking and that the printed descriptions match the actual cards. I organized these tests into a small internal "test sheet" that the group can reuse later as regression tests.

Week-5 (9 Mar – 15 Mar):

After the Expectiminimax module was integrated, I participated in several test games against the AI. I noted situations where the bot's decisions appeared reasonable and cases where it looked too aggressive or too passive. I began drafting descriptive text for the documentation that explains the user interaction flow, sample game scenarios, and observed behaviors of the AI. I also suggested minor message improvements to make the game easier to follow for new users.

Current Status:

My main contributions are in user- oriented requirements, interface design, scenario- based testing, and documentation of behavior.

In the next phase I plan to help design structured experiments for evaluating the AI, assist in writing both the 2- page update report and 8- page final report in proper format, and work on polishing the slides for the class presentations.